

Winding Tree

Laying down the foundations of Winding Tree decentralized travel marketplace



Mathieu Tahon · Following

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At Winding Tree we are on a mission to completely change the way travel is being distributed, making it permission-less, transparent, and inclusive.

During this journey, we have identified bottlenecks in the legacy travel distribution and we have created an infrastructure that we believe will allow sustainable growth by allowing buyers and suppliers to transact with one another without asking the permission of any third party, including the Winding Tree team itself.

We are big believers in decentralized technology and open-source software, and in this spirit, we have built the layers and tools for this future-proof and secure marketplace. The technologies we use are cutting-edge, most of them did not exist only 10 years ago, which contrasts with the software being used in many legacy travel companies, sometimes dating from the 70s.

The aim of this article is to provide an overview of the technical components used in the Winding Tree marketplace.

The layers of this marketplace are as follow (the "big picture" summary is at the end of the article):

1. Transaction Layer: REST/JSON API with an OAuth twist
2. Trust & Ownership Layer: Ethereum Blockchain
3. Storage Layer: InterPlanetary File System (IPFS) and W3C DID
4. Caching & Indexing: GraphQL powered by The Graph Protocol

Transaction Layer: REST/JSON API with an OAuth twist

The travel industry processes huge amounts of data and the volume of transactions performed for shopping and booking is mind-blowing. To support this volume, we use the real-time and battle-tested HTTP protocol which is pretty common in digital commerce, and a modern REST/JSON presentation interface documented in an OpenAPI specification to ease the developer experience.

Because the Winding Tree marketplace is decentralized, any travel supplier can expose an HTTP Service endpoint for buyers to shop and order its products and services. This is a peer-to-peer model, with no reliance on a central provider.

Having no centralized party in the marketplace is great, but how is it possible to know who is the other party in the transaction? How to authenticate and verify the other party in a scalable manner?

A classic method for authentication in modern applications is to leverage to OAuth 2 authentication mechanism, using typically a *Bearer authentication token* in the headers of the HTTP request. Such Bearer is normally obtained from a separate API endpoint using credentials or API key provided from a web portal or manually by the supplier.

In the Winding Tree marketplace, we use a Bearer token as well, but instead of getting such a token from an API endpoint, there is a tweak: the token is self-generated by the client itself. This is made possible with the design of JSON Web Tokens (JWT), indicating as a signing key an authentication key owned by the client's organization. This leads us to the next underlying layer: the Ethereum blockchain.

Trust & Ownership Layer: Ethereum Blockchain

Blockchains in general are a revolution in digital commerce as they introduce a concept of digital and global ownership, in other terms a Distributed Ledger. In a blockchain like Bitcoin, it is possible to own digital currencies and transfer it to another party. With the Ethereum blockchain, the ownership is no longer limited to currencies. It is possible to own digital arts, collectibles, and other assets.

In the Winding Tree marketplace, travel businesses can *own* a digital organization representing them: we call it an ORGiD. Having a digital representation of the ownership of a travel business creates endless opportunities and a few challenges.

The main challenge is Trust: how do I know this digital organization is actually the same as the physical one and not a scam? Is the other business from a country with an embargo or involved in illegal activities?

In the Winding Tree marketplace, we have introduced different mechanisms for Know Your Business (KYB) procedures:

- A naive way which is always possible is for participants to make a list of trusted peers, allowing business one-by-one with offline communication of their identifiers (ORGiD value)
- This process can be partially automated by making a two-way link with other digital presence: starting from the domain name (DNS) of a company website, SSL EV certificates, but also social media accounts. If

you trust a company website is at *airline.com*, probably you can trust their ORGiD identifier is correct if it is published at *airline.com/orgid.txt*

- Finally, we are rolling out in partnership with Kleros the concept of *arbitrable directories* where organizations are curated by a decentralized justice court. Details of such integration deserve a separate article.

All the trust and descriptive data associated with an organization can represent quite some information. This storage comes at a cost, and in Blockchains it is particularly expensive. Therefore in the Winding Tree marketplace, the storage of extensive company information is stored off-chain, while only a digital fingerprint (or hash) of this data is stored along with the ORGiD identifier. This is happening in the next layer: storage.

Storage Layer: InterPlanetary File System (IPFS) and W3C DID

The InterPlanetary File System (IPFS) is a file distribution protocol that gained traction in decentralized applications. It is highly efficient it removes the correlation between addressing and storage location, allowing optimizations like de-duplication, caching, and network path optimizations.

In IPFS all files are addressed by their digital fingerprint (or hash), making it a perfect fit for the need of file integrity in the Winding Tree marketplace. Indeed, because a fingerprint is stored along with the ORGiD in the Ethereum smart contract, there is a bijection to its associated document in the IPFS network. This means that it is impossible to modify the organization extended details, as any modification would be a different file!

For the file structure, we have implemented the World Wide Web Consortium (W3C) proposed standard for decentralized identities: W3C DID. This standard is based on JSON and, among other things, structure the way public keys and service endpoints are defined. We have extended this standard to include additional details required for travel organizations. By using this standard, we are ensuring all parties understand how to retrieve

the details of another party and we facilitate the communication with third party projects and protocols.

Caching & Indexing: GraphQL powered by The Graph Protocol

All the above layers are the base of the Winding Tree marketplace and together are providing the foundations of an open-secure decentralized travel marketplace.

However, with so many layers it can become technically complex for travel businesses to be part of the marketplace. For travel suppliers, it means being able to query an Ethereum node and IPFS node for the details of travel buyers making requests. For travel buyers, it means monitoring the Ethereum blockchain for new participants to connect to.

To facilitate the discovery and authentication process, we are providing a subgraph for The Graph protocol, a project that allows creating a meaningful index from events happening on the Ethereum blockchain, enriched with IPFS data. It allows participants to host (or source) their own GraphQL endpoint to find new organizations.

Below is an example GraphQL query/reply which can be found on the hosted version of The Graph here.

Conclusion

The Travel Industry is still powered by systems built in the infancy of the Internet but still very much in use today because of a high degree of centralization.

Re-platforming an industry is a huge undertaking, but at Winding Tree we believe technologies are present today to create an inclusive marketplace where direct connections are the norm rather than the exception.

The below diagram summarizes our vision of the different technologies and layers orchestrating the future travel marketplace:

Winding Tree marketplace: the big picture

Interested in joining the marketplace? Create your organization at <https://marketplace.windingtree.com> then implement the protocol using our developer documentation at <https://developers.windingtree.com>. Stuck at some point? Reach out at partners@windingtree.com.

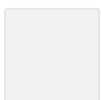
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Decentralized Travel Distribution

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Responses (2)



Maksim Izmaylov

What are your thoughts?



Mathieu Tahon

Author

Nov 9, 2020



Hi Martin!

Yes your understanding is correct, but it can be generalized a bit more outside of the Travel Agency to Airline use case: for example it can accomodate cases where airlines buy from hotels (eg: for their staff or disruption), or directly... [more](#)



[Reply](#)



Martin Spinner

Nov 9, 2020



If I understand correctly, the functional goal of this system is to authenticate the travel agency to the airline in a decentralized way. What are the advantages it brings compared to a centralized authentication system (centralized marketplace) ?



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